



Principal = ₹ 1,00,000

$R\% = 10\%$

$T = 1\text{yr}$

* Simple Interest

$$= 10\% \times 1,00,000$$

$$= \frac{10}{100} \times 1,00,000$$

$$= 10,000 \text{ [Extra Money]}$$

$$\text{Principal} = ₹ 1,00,000$$

$$R\% = 10\%$$

$$T = 2 \text{ yr}$$

SI $\Rightarrow$

$$1 \text{ yr} \rightarrow 10 \text{ k} = 1 \times 10 \text{ k} //$$

$$\underline{2 \text{ yr}} \rightarrow 10 \text{ k} + 10 \text{ k} = \underline{20 \text{ k}} = 2 \times 10 \text{ k} //$$

$$\underline{3 \text{ yr}} \rightarrow \underline{30 \text{ k}} = 3 \times 10 \text{ k} //$$

⋮

$$\underline{\underline{T \text{ yrs}}} \rightarrow T \times 10 \text{ k} //$$

$$\underline{T_{\text{yrs}}} \rightarrow T \times 10^k //$$

$$= T \times R\% \times P$$

$$SI = T \times R\% \times P$$



$$Q] \quad P = 1,00,000 \text{ ₹}$$

$$\%R = 10\% \text{ [Annually]}$$

$$T = 2 \text{ yrs}$$

MI Final Amount = ?

* By Default $\rightarrow$ [S.I]

$$\begin{aligned} \textcircled{1} \quad SI &= T \cdot R\% \cdot P = 2 \times \frac{10}{100} \times 1L \\ &= 20K // \end{aligned}$$

. ② $A = P + SI$

$$= 1L + 20k$$

$$= 1,20,000 \text{ ₹} //$$

(M2)

$$A = P + SI$$

$$= P + [TR\% \cdot P]$$

Final
Amount

$$A = P \left[1 + \frac{TR}{100} \right]$$

$$A = 1L \left[1 + \frac{2 \times 10}{100} \right]$$

$$A = 1,20,000 \text{ ₹,,}$$

Q]

$$P = 1,00,000 \text{ ₹}$$

$$\%R = \underline{10\%} \quad [\text{Compounded Annually}]$$

$$T = 2 \text{ yrs}$$

final Amount = ?

(M1)

1st yr :

$$A_1 = P \left[1 + \frac{TR}{100} \right]$$

$$A_1 = 1L \left[1 + \frac{1 \times 10}{100} \right] = 1.1L //$$

✓ nd
2yr :

$$A = A_1 \left[1 + \frac{TR}{100} \right]$$

$$= 1.1 L \left[1 + \frac{(1)(10)}{100} \right]$$

$$A = 1,21,000 \text{ ₹} //$$

$$\boxed{T = 24 \text{ rs}}$$

$$A = A_1 \left[1 + \frac{TR}{1\omega} \right]$$

$$A_1 = P \left[1 + \frac{TR}{1\omega} \right]$$

$$\Rightarrow A = P \left[1 + \frac{1 \cdot R}{1\omega} \right] \left[1 + \frac{1 \cdot R}{1\omega} \right]$$

C.I

% Rate of Interest—

$$A = P \cdot \left[1 + \frac{R}{100} \right]^n$$

↓
n years

Shrenik Jain — Study Simplified

Final Amount

CI

$$A = P \cdot \left[1 + \frac{R}{100} \right]^n$$

SI

$$A = P \left[1 + \frac{TR}{100} \right]$$

$$\{ SI = CI \text{ when } T = n = 1 \text{ yr} \}$$

Find Compound interest on Rs 10,000 for 2 years at a
Rate of interest=10% per annum compounded ~~half yearly~~.
Find the total amount due at the end of 2 years

* If No half yr compounding

① CI

② $A = P \left[1 + \frac{R}{100} \right]^n$

$$= 10,000 \left[1 + \frac{10}{100} \right]^2$$

$$= 12,100 \text{ ₹ //$$

Find Compound interest on Rs 10,000 for 2 years at a
Rate of interest=10% per annum compounded half yearly.
Find the total amount due at the end of 2 years

(M1)

$$\% R = 10\% \quad (1 \text{ yr})$$

$$\% R \text{ for } \frac{1}{2} \text{ yr} = \frac{10}{2} \% = 5\%$$

$$1 \text{ yr} \rightarrow 2 \text{ Half yrs}$$

$$2 \text{ yrs} \rightarrow 4 \text{ Half yrs}$$

2n (1/2yr)

Half yr

P

Interest -

$(R/2)\%$

1/2yr

(Amount due)

1st

10,000

500

10500 ✓

2nd

10,500

525

11025

3rd

11025

551.25

11576.25

4th

11576.25

578.81

12155.06 //

(M2)

$$A = P \left[1 + \frac{R/2}{100} \right]^{2n}$$

$\frac{1}{2}$ yr $\Rightarrow \% R$

How much

$\frac{1}{2}$ yrs //

Compounded

$\frac{1}{2}$ yrly

Compounded
annually

$$\Rightarrow A = P \left[1 + \frac{R}{100} \right]^n$$

Compounded
half yrly

$$\Rightarrow A = P \left[1 + \frac{R/2}{100} \right]^{2n}$$

Compounded
quarterly

$$\Rightarrow A = P \left[1 + \frac{R/4}{100} \right]^{4n}$$

Find Compound interest on Rs $P = 10,000$ for $n = 2$ years at a Rate of interest $r = 10\%$ per annum compounded half yearly. Find the total amount due at the end of 2 years

M2

[Trick]

$$A = P \left[1 + \frac{R/2}{100} \right]^{2n}$$

$$= 10,000 \left[1 + \frac{5}{100} \right]^{2 \times 2}$$

$$= 12155.0625 //$$

The CI on Rs 5000 for 3 years at 8% for first year, 10% for second year and 12% for third year will be

- (A) Rs 1750
- ☒ (B) Rs 1652.80
- (C) Rs 1575
- (D) Rs 1685.20

① CI

$$\textcircled{2} A = P \left[1 + \frac{R_1}{100} \right] \left[1 + \frac{R_2}{100} \right] \left[1 + \frac{R_3}{100} \right]$$

$$= 5000 \left[1 + \frac{8}{100} \right] \left[1 + \frac{10}{100} \right] \left[1 + \frac{12}{100} \right]$$

$$A = 6652.8 \text{ ₹}$$

$$A = 6652.8 \text{ ₹}$$

Extra money ???

$$= A - P$$

$$= 6652.8 - 5000$$

$$= 1652.8 \text{ ₹} //$$

Leila aspires to buy a car worth Rs 10,00,000 after 5 years. What is the minimum amount in Rs that she should deposit now in a bank which offers 10% annual rate of interest, if the interest was compounded annually?

(A) 5,00,000

✓ (B) 6,21,000

(C) 6,66,667

(D) 7,50,000

[GATE 2018 EC]

(1) CI

$$(2) A = P \left[1 + \frac{R}{100} \right]^T$$

$$10^6 = P \left[1 + \frac{10}{100} \right]^5$$

$$P \approx 6.21 \text{ lakhs}$$

Q] CI on ₹ 1600 at 25% per annum of 2 yrs compounded annually?

Solⁿ:

① CI

$$\begin{aligned} \text{② } A &= P \left[1 + \frac{R}{100} \right]^T \\ &= 1600 \left[1 + \frac{25}{100} \right]^2 \end{aligned}$$

$$A = 2500 \text{ ₹}$$

③

$$CI = A - P$$

$$= 2500 - 1600$$

$$= 900 \text{ ₹} //$$

The population of a new city is 5 million and is growing at 20 % annually. How many years would it take to double at this growth rate?

☒ (A) 3-4 years

☐ (B) 4-5 years

(C) 5-6 years

(D) 6-7 years

[GATE 2014- CE/IN/PI]

$$P = 5 \text{ million}$$

(M1)

$$\% R = 20\% \quad \checkmark$$

1st 4r

$$A_1 = P \left[1 + \frac{20}{100} \right]$$

$$= 5 \text{ m} \left[\frac{6}{5} \right] = \underline{\underline{6 \text{ m}}}$$

2nd 4r

$$A_2 = 6 \text{ m} \left[1 + \frac{20}{100} \right] = 7.2 \text{ m}$$

∴ 3rd yr

$$A_3 = 7.2m \left[1 + \frac{20}{100} \right] = 8.64m$$

4th yr

$$A_4 = 8.64m \left[1 + \frac{20}{100} \right] = 10.368m //$$

$$A = P \left[1 + \frac{R}{100} \right]^T$$

(M2)

[Comp.
Interest]

$$10m = 5m \left[1 + \frac{20}{100} \right]^T$$

$$2 = \left(\frac{6}{5} \right)^T$$

$$\ln 2 = \ln \left(\frac{6}{5} \right)^T$$

$$\ln 2 = \ln \left(\frac{6}{5} \right)^T$$

$$\ln 2 = T \ln \left(\frac{6}{5} \right)$$

$$T = \frac{\ln 2}{\ln \left(\frac{6}{5} \right)}$$

$$= 3.8 \text{ yrs } //$$

A sum of money placed at compounded interest doubles ✓
itself in 5 years. ✓ It will amount to eight times itself at the
same rate of interest in

(A) 7 years

(B) 10 years

✓ (C) 15 years

(D) 20 years

①

P

$$A = 2P$$

$$T = 5 \text{ yr}$$

$$A = P \left[1 + \frac{R}{100} \right]^T$$

$$2P = P \left[1 + \frac{R}{100} \right]^5$$

②

$$A = 8P$$

$$\cancel{8P} = \cancel{P} \left[1 + \frac{R}{100} \right]^T$$

$$\{ \cancel{2P} \}^3 = \{ \cancel{P} \left[1 + \frac{R}{100} \right]^5 \}^3$$

$$\left[1 + \frac{R}{100} \right]^T = \left[1 + \frac{R}{100} \right]^{15}$$

$$T = 15 \text{ yrs}$$

$$(a^m)^n = a^{m \cdot n}$$

A sum of Rs 12000 is deposited into SBI Bank for 3 years.
If the bank is providing 5% rate then calculate the amount after the maturity period.

Solⁿ: ✓ ① SI

② $A = P \left[1 + \frac{RT}{100} \right]$

$$= 12000 \left[1 + \frac{5 \times 3}{100} \right]$$

$$= 13800 //$$